LECTURE 16

1. Introduction to the Immune System

This slide establishes the two main branches of the immune system and their key differences[1].

The Immune System's Goal

The immune system works through **multiple layers of protection** to **prevent** and **eliminate Pathogens** (such as bacteria, fungi, and viruses)[2].

Feature	**Innate Immunity**	**Adaptive Immunity**		
Organisms	Found in **all animals and plants**[5555].	Found **only in vertebrates**[6666].		
Response Time		**Rapid response**[7].		**Slower response**[8].
Recognition	Recognition of **traits shared by broad ranges of pathogens**, using a small set of receptors[9].	Recognition of **traits specific to particular pathogens**, using a vast array of receptors[10].		

Defenses		**Barrier defenses** (Skin, Mucous membranes, Secretions) and **Internal defenses** (Phagocytic cells, Natural killer cells, Antimicrobial proteins, Inflammatory response)[11].		**Humoral response** (**Antibodies** defend against infection in body fluids) and **Cell-mediated response** (Cytotoxic cells defend against infection in body cells)[12].

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2 & 3. Haematopoiesis (Origin of Blood Cells)

These slides illustrate the process of **Haematopoiesis**, the formation of all blood cell types, including immune cells, and identify where this process occurs in the body.

Haematopoietic Lineage [13]

All blood cells originate from **Hematopoietic stem/Progenitor cells**[14]. This lineage splits into two major pathways:

1. **Common Myeloid Progenitor cells**[15]: This lineage gives rise to most innate immune cells and red blood cells.
    - **Megakaryocyte-Erythrocyte progenitor cells** [16]lead to **Platelets (Thrombocytes)** and **Erythrocytes (RBCs)**[17].
    - **Granulocyte-Monocyte progenitor cells** [18]lead to the **Myeloid Lineage** [19] cells:
        - **Granulocytes**: **Basophils**, **Neutrophils**, and **Eosinophils**[20202020].
        - **Agranulocytes**: **Monocytes**, which mature into **Macrophages** and **Dendritic cells**[21212121].
2. **Common Lymphoid progenitor cells**[22]: This lineage gives rise to the key cells of the adaptive immune system and certain innate cells.

- ○ **Lymphoblasts** [23] mature into **Lymphocytes** [24] (**B and T Lymphocytes**) and **NK cells** (Natural Killer cells)[25].

Where Haematopoiesis Occurs

In adults, haematopoiesis primarily occurs in the bone marrow of the **Axial skeleton**[26]:

- **Skull** [27]
- **Sternum** [28]
- **Vertebrae** [29]
- **Iliac crest** (hip bone) [30] - This is the usual site for **bone marrow biopsies**[31].
- The **Proximal end of large bones**[32]. (The location is **Different in children** [33]).

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4. Circulatory System vs. Lymphatic System

This slide shows the major components of the Circulatory system (red and blue vessels) and the **Lymphatic System** (labeled organs and vessels).

- The Lymphatic System is a network of tissues and organs that help rid the body of toxins, waste and other unwanted materials. Its main functions are **Fluid balance**, **Immune responses**, and **Fat absorption**[34].
- Key immune-related organs of the lymphatic system include: **Thymus gland**, **Spleen**, **Tonsils**, **Adenoids**, **Peyer's patches** (in the small intestine), and **Lymph nodes** (e.g., cervical, axillary, inguinal)[3535353535353535353535353535353535353535353535353535353535
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353535]. **Bone marrow** is also part of the lymphatic system where lymphocytes mature[36].

5, 6, 7, 8 & 9. Adaptive Immunity: B Cells and T Cells

These slides detail the cellular players (**B cells** and **T cells**) and the molecular tools (**antigen receptors**) of **Adaptive or Acquired Immunity**[37].

Lymphocyte Maturation and Antigens

- Mature **B cells** and **T cells** originate from stem cells in the **bone marrow**[38].
- **B cells** mature in the **Bone marrow**[39].
- **T cells** migrate to and mature in the **Thymus**[40].
- An **Antigen** is any substance that elicits a B or T cell response[41]. The small, accessible portion of an antigen that binds to a receptor is called an **Epitope**[42424242].

The B Cell Antigen Receptor and Antibody

- Recognition occurs when a **B cell** binds to an **Antigen** via a **B cell antigen receptor** (a protein)[43434343].
- **Structure**[44]:
  - Composed of **two identical heavy chains** and **two identical light chains**[4545].
  - Chains are linked by **Disulfide bridges**[46464646].
  - Has **two identical antigen binding sites**[47474747].
  - Each chain has a **Constant (C) region** (sequence varies little) [48484848]and a **Variable (V) region** (sequence varies extensively)[49494949]. The V regions form the binding site[50].
- **Antibody Production**: Binding of a receptor to an antigen activates the B cell[51].

  Activated B cells produce a secreted form of the receptor called an **Antibody**[52]. Antibodies have the **same Y-shaped structure** as B cell receptors but are secreted and can bind to antigens on pathogens or free in body fluids[53535353].

The T Cell Antigen Receptor and Antigen Presentation

- **T Cell Receptor Structure**[54]:

    - Consists of **two different polypeptide chains**: an **α chain** and a **β chain**, linked by a **Disulfide bridge**[555555555555555555].
    - Has a **single antigen-binding site**[56565656].
    - Contains **Variable (V)** and **Constant (C) regions**[57575757].
- **Unique Recognition**: T cell receptors bind **only to fragments of antigens displayed** (or presented) on the surface of host cells[58].
- **Antigen Presentation**[59]:
    - Pathogen is taken in or infects a **Host cell**[60].
    - Enzymes cleave the antigen into smaller **Antigen fragments**[61616161].
    - The fragment binds to an **MHC molecule** (Major Histocompatibility Complex, the host display protein) inside the cell[626262626262626262].
    - The MHC molecule and bound fragment move to the cell surface, resulting in **Antigen presentation**, which is "advertising the fact that a host cell contains a foreign substance"[63].

10, 11 & 12. Clonal Selection and Immunological Memory

These slides explain how the adaptive immune system generates a massive response and retains memory.

Clonal Selection

- The body produces millions of different B cell and T cell receptors (over 10^6 and 10^7, respectively)[64].
- Only **one B cell** whose receptor matches the **Antigen** is **selected** and **activated**[65656565].
- Once activated, the B cell undergoes multiple cell divisions (**proliferation**), creating a **clone—**a population of **identical cells** bearing the same receptor[66666666].

Effector Cells and Memory

The clone differentiates into two types of daughter cells[67]:

1. **Effector cells** (short-lived cells that act immediately)[68]:
    - The effector form of B cells are **Plasma cells**, which **secrete antibodies**[696969696969696969].
    - The effector forms of T cells are **Helper T cells** and **Cytotoxic T cells**[70].
2. **Memory cells** (long-lived cells)[71]:
    - These cells remain and can **respond rapidly upon subsequent exposure to the same antigen**[72727272].

- This mechanism is the basis of **Immunological memory**, which is responsible for **long-term protection**[73].

Primary vs. Secondary Immune Response

The graph illustrates the time course and magnitude of the immune response:

- **Primary Immune Response** (Exposure to Antigen A for the first time): It takes about 7-14 days to generate a detectable antibody concentration, which peaks around day 21[74747474].
- **Secondary Immune Response** (Later Exposure to Antigen A): The response is **much faster** (starting almost immediately), **stronger** (reaching $\approx 10^4$ arbitrary units), and **longer-lasting**[7575757575].
- **Specificity**: When exposed to both **Antigens A and B** simultaneously, the response to **A is secondary**, but the response to **B is primary**, confirming that immunological memory is **specific to an antigen and does not affect responses to other antigens**[767676767676767676].

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13. How Antibodies Function

Antibodies **do not actually kill pathogens**[77]. By binding to antigens, they interfere with pathogen activity or mark them for inactivation/destruction[78].

1. **Neutralization**: Antibodies bound to antigens on the surface of a **Virus neutralize it by blocking its ability to bind to a host cell**[79797979].

2. **Opsonization** (not pictured, but described in context): Binding of antibodies to antigens on the surface of **Bacteria** promotes **Phagocytosis** by **Macrophages** and **Neutrophils**[80].
3. **Activation of Complement System and Pore Formation**:
    - Binding of antibodies to antigens on the surface of a **Foreign cell** (e.g., bacteria) **activates the complement system**[81].
    - Following activation, the **Complement proteins** form a **Membrane Attack Complex**[82].
    - This complex forms **Pores** in the foreign cell's membrane, allowing **water and ions to rush in**[83838383838383].
    - The cell **swells and eventually lyses**[84].

1. Clonal Selection and Memory Formation

This figure explains how the immune system rapidly generates a powerful, antigen-specific response and creates long-term memory cells.

Step 1: Antigen Binding and Selection

- The immune system maintains a large, diverse pool of **B cells that differ in antigen specificity**[1].
- These B cells have unique **Antigen receptors** on their surface[2].
- **Antigens bind to the antigen receptors of only one of the B cells shown**—the one whose receptor perfectly matches the antigen's epitope[3]. This selection process is called **clonal selection**.

Step 2: Proliferation (Cloning)

- **The selected B cell proliferates**, undergoing multiple cell divisions to form a large **clone of identical cells**[4].
- All cells in this clone bear the exact same receptors for the antigen that initially triggered the response[5].

Step 3: Differentiation into Effector and Memory Cells

- The clone differentiates into two populations of daughter cells:
    - **Plasma cells (Effector cells):** These are short-lived cells that **secrete antibodies specific for the antigen** into the body fluids to fight the current infection[6].
    - **Memory cells:** These are **long-lived cells** that remain in the body[7]. They enable the immune system to **respond rapidly upon subsequent exposure to the same antigen**[8].

This process ensures that the body mounts an immediate defense while preparing for much faster and stronger responses to any future infections by the same pathogen.

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2. Primary vs. Secondary Immune Response

This graph is a quantitative representation of **immunological memory**, showing the antibody concentration in the blood over time following exposure to two different antigens (A and B).

Primary Immune Response (Antigen A, Day 0)

- **Initial Exposure:** The body is exposed to **Antigen A** on Day 0[9].
- **Slow Start:** There is a delay (lag phase) before antibodies are produced (about 7 days)[10].
- **Magnitude and Duration:** The **Primary immune response to antigen A produces antibodies to A**[11]. This response is relatively **small** (peaking around 10^1 to 10^2 arbitrary units) and **short-lived**, declining around Day 28[12]. This initial response is slow because the specific B cells must first be selected, cloned, and differentiated (as shown in the previous figure).

Secondary Immune Response (Antigen A & B, Day 28)

- **Second Exposure:** The body is exposed to **antigens A and B** simultaneously on Day 28[13].
- **Response to Antigen A (Secondary Response):**

- **Speed and Magnitude:** Because of the **memory cells** generated during the primary response, the **secondary immune response to antigen A** is **immediate, much faster**, and **massively stronger**[14].
- The antibody concentration for A soars to $\approx 10^4$ arbitrary units, a concentration 100 to 1,000 times greater than the primary response[15].

- **Response to Antigen B (Primary Response):**
  - **Specificity:** The body simultaneously mounts a **primary immune response to antigen B**[16].
  - This response has the same **slow start, small magnitude** (peaking below 10^2 units), and is **short-lived**, identical to the original primary response to Antigen A[17].

Key Conclusion

The data conclusively demonstrates that **Immunological memory is specific to an antigen and does not affect responses to other antigens**[18]. The prior exposure to A only boosted the response to A, while the response to the new antigen B had to start from scratch.